TUF+ DSA Sheet

Topic	Problem Name	Completed	Revisit	Revision Count	Notes
Basics	Java Collections/STL				
	Maths				
	Count all Digits of a Number				
	Count number of odd digits in a number				
	Reverse a number				
	Palindrome Number				
	Return the Largest Digit in a Number				
	Factorial of a given number				
	Check if the Number is Armstrong				
	Check for Perfect Number				
	Check for Prime Number				
	Count of Prime Numbers till N				
	GCD of Two Numbers				
	LCM of two numbers				
	Divisors of a Number				
	Arrays				
	Sum of Array Elements				
	Count of odd numbers in array				
	Check if the Array is Sorted I				
	Reverse an array				
	Hashing				
	Highest Occurring Element in an Array				
	Second Highest Occurring Element				
	Sum of Highest and Lowest Frequency				
	Strings				
	Reverse a String II				
	Palindrome Check				
	Largest Odd Number in a String				
	Longest Common Prefix				
	Isomorphic String				
	Rotate String				
	Valid Anagram				
	Sort Characters by Frequency				
	Recursion				
	Recursion Theory				
	Recursion Concepts with Parameters				
	Sum of First N Numbers				
	Factorial of a Given Number				
	Sum of Array Elements				
	Reverse a String I				
	Check if String is Palindrome or Not				
	Check if a Number is Prime or Not				
	Reverse an array				
	Check if the Array is Sorted II				
	Sum of Digits in a Given Number				
	Fibonacci Number				
Sorting	Selection Sort				
	Bubble Sort				
	Insertion Sorting				
	Merge Sorting				
	Quick Sorting				
Arrays	Linear Search				
	Largest Element				
	Second Largest Element				
	Maximum Consecutive Ones				
	Left Rotate Array by One				
	Left Rotate Array by K Places				

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Arrays	Move Zeros to End				
	Remove duplicates from sorted array				
	Find missing number				
	Union of two sorted arrays				
	Intersection of two sorted arrays				
Arrays Medium	Majority Element-I				
	Leaders in an Array				
	Rearrange array elements by sign				
	Print the matrix in spiral manner				
	Pascal's Triangle I				
	Pascal's Triangle II				
	Pascal's Triangle III				
	Rotate matrix by 90 degrees				
	Two Sum				
	3 Sum				
	4 Sum				
	Sort an array of 0's 1's and 2's				
	Kadane's Algorithm				
	Next Permutation				
Arrays Hard	Majority Element-II				
	Find the repeating and missing number				
	Count Inversions				
	Reverse Pairs				
	Maximum Product Subarray in an Array				
	Merge two sorted arrays without extra space				
Hashing	Longest Consecutive Sequence in an Array				
	Longest subarray with sum K				
	Count subarrays with given sum				
	Count subarrays with given xor K				
Binary Search	Search X in sorted array				
	Lower Bound				
	Upper Bound				
	Search insert position				
	Floor and Ceil in Sorted Array				
	First and last occurrence				
	Search in rotated sorted array-I				
	Search in rotated sorted array-II				
	Find minimum in Rotated Sorted Array				
	Find out how many times the array is rotated				
	Single element in sorted array				
	Find square root of a number				
	Find Nth root of a number				
	Find the smallest divisor				
	Koko eating bananas				
	Minimum days to make M bouquets				
	Aggressive Cows				
	Book Allocation Problem				
	Find peak element				
	Median of 2 sorted arrays				
	Kth element of 2 sorted arrays				
	Minimize Max Distance to Gas Station				
	Split array - largest sum				
BS(2D Arrays)	Find row with maximum 1's				
	Search in a 2D matrix				
	Search in 2D matrix - II				
	Find Peak Element - II				
	Matrix Median				
Recursion	Pow(x,n)				
	Generate Parentheses				
	Power Set				

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Recursion	Check if there exists a subsequence with sum K				
	Count all subsequences with sum K				
Recursion Medium	Combination Sum				
	Combination Sum II				
	Subsets I				
	Subsets II				
	Combination Sum III				
Recursion Hard	Letter Combinations of a Phone Number				
	Palindrome partitioning				
	Word Search				
	N Queen				
	Rat in a Maze				
	M Coloring Problem				
	Sudoku Solver				
Single linked list	Introduction to Singly LinkedList				
	Traversal in Linked List				
	Deletion in Linked List				
	Insertion in Linked List				
	Deletion of the head of LL				
	Deletion of the tail of Linked List				
	Deletion of the Kth element of Linked List				
	Delete the element with value X				
	Insertion at the head of Linked List				
	Insertion at the tail of Linked List				
	Insertion at the Kth position of Linked List				
	Insertion before the value X in Linked List				
Double Linked List	Introduction to Doubly LL				
	Deletion in Doubly LL				
	Insertion in DLL				
	Convert Array to Doubly Linked List				
	Delete head of Doubly Linked List				
	Delete Tail of Doubly Linked List				
	Delete Kth Element of Doubly Linked List				
	Removing given node in Doubly Linked List				
	Insert node before head in Doubly Linked List				
	Insert node before tail in Doubly Linked List				
	Insert node before (kth node) in Doubly Linked List				
	Insert before given node in Doubly Linked List				
Logic Building	Add two numbers in Linked List				
	Segregate odd and even nodes in Linked List				
	Sort a Linked List of 0's 1's and 2's				
	Remove Nth node from the back of the LL				
	Reverse a LL				
Linked list Medium	Add one to a number represented by Linked List				
	Find Middle of Linked List				
	Delete the middle node in LL				
	Check if LL is palindrome or not				
	Find the intersection point of Y LL				
	Detect a loop in Linked List				
	Find the starting point in LL				
	Length of loop in LL				
Linked list (Hard)	Reverse LL in group of given size K				
	Rotate a Linked List				
	Merge two Sorted Lists				
	Flattening of LL				
	Sort LL				
	Clone a LL with random and next pointer				
	Delete all occurrences of a key in DLL				
	Remove duplicated from sorted DLL				
Bit Manipulation	Minimum Bit Flips to Convert Number				

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Bit Manipulation	Single Number - I				
	Single Number - II				
	Single Number - III				
	Divide two numbers without multiplication and division				
	Power Set Bit Manipulation				
	XOR of numbers in a given range				
Greedy Algorithms	Assign Cookies				
	Lemonade Change				
	Jump Game - I				
	Shortest Job First				
	Job sequencing Problem				
	N meetings in one room				
	Non-overlapping Intervals				
	Insert Interval				
	Minimum number of platforms required for a railway				
Gree dy Hard	Valid Paranthesis Checker				
	Candy				
Sliding Window /2Pointer	Maximum Points You Can Obtain from Cards				
	Longest Substring Without Repeating Characters				
	Max Consecutive Ones III				
	Fruit Into Baskets				
	Longest Substring With At Most K Distinct Characters				
	Longest Repeating Character Replacement				
	Minimum Window Substring				
	Number of Substrings Containing All Three Characters				
	Binary Subarrays With Sum				
	Count number of Nice subarrays				
Stack/Queue Implementation	Implementation using different DS				
	Implement Stack using Arrays				
	Implement Queue using Arrays				
	Implement Stack using Queue				
	Implement Queue using Stack				
	Implement stack using Linkedlist				
	Implement queue using Linkedlist				
	Balanced Paranthesis				
Stack/Queue FAQs	Next Greater Element				
	Next Greater Element - 2				
	Asteroid Collision				
	Sum of Subarray Minimums				
	Sum of Subarray Ranges				
	Remove K Digits				
	Implement Min Stack				
	Sliding Window Maximum				
	Trapping Rainwater				
	Largest rectangle in a histogram				
	Maximum Rectangles				
	Stock span problem				
	Celebrity Problem				
	LRU Cache				
	LFU Cache				
Binary Tree Traversal	Inorder Traversal				
	Preorder Traversal				
	Postorder Traversal				
	Level Order Traversal				
	Pre, Post, Inorder in one traversal				
Binary Tree	Maximum Depth in BT				

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Binary Tree (Medium)	Check if two trees are identical or not				
	Check for balanced binary tree				
	Diameter of Binary Tree				
	Maximum path sum				
	Check for symmetrical BTs				
	Zig Zag or Spiral Traversal				
	Boundary Traversal				
	Vertical Order Traversal				
	Top View of BT				
	Bottom view of BT				
	Right/Left View of BT				
	Print root to node path in BT				
	LCA in BT				
	Maximum Width of BT				
	Print all nodes at a distance of K in BT				
	Minimum time taken to burn the BT from a given Node				
	Count total nodes in a complete BT				
Construction Problems	Requirements needed to construct a unique BT				
	Construct a BT from Preorder and Inorder				
	Construct a BT from Postorder and Inorder				
	Serialize and De-serialize BT				
Travel In Constant Space	Morris Inorder Traversal				
	Morris Preorder Traversal				
Binary Search Tree	Search in BST				
	Floor and Ceil in a BST				
	Insert a given node in BST				
	Delete a node in BST				
	Kth Smallest and Largest element in BST				
	Check if a tree is a BST or not				
	LCA in BST				
	Construct a BST from a preorder traversal				
	Inorder successor and predecessor in BST				
	BST iterator				
	Two sum in BST				
	Correct BST with two nodes swapped				
	Largest BST in Binary Tree				
Heaps	Heapify Algorithm				
	Build heap from a given Array				
	Implement Min Heap				
	Implement Max Heap				
	Check if an array represents a min heap				
	Convert Min Heap to Max Heap				
	Heap Sort				
	K-th Largest element in an array				
	Kth largest element in a stream of running integers				
Graphs Traversal	Traversal Techniques				
	Connected Components				
	Number of provinces				
	Number of islands				
	Flood fill algorithm				
	Number of enclaves				
	Rotten Oranges				
	Distance of nearest cell having one				
	Surrounded Regions				
	Number of distinct islands				
Graphs Cycles	Detect a cycle in an undirected graph				
	Bipartite graph				
	Topological sort or Kahn's algorithm				
	Detect a cycle in a directed graph				

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Graphs Hard	Find eventual safe states				
	Course Schedule I				
	Course Schedule II				
	Alien Dictionary				
Graphs Hard	Shortest path in DAG				
	Shortest path in undirected graph with unit weights				
	Word ladder I				
	Word ladder II				
Shortest Path Algo	Dijkstra's algorithm				
	Print Shortest Path				
	Shortest Distance in a Binary Maze				
	Path with minimum effort				
	Cheapest flight within K stops				
	Minimum multiplications to reach end				
	Number of ways to arrive at destination				
	Bellman ford algorithm				
	Floyd warshall algorithm				
	Find the city with the smallest number of neighbors				
Minimum Spanning Tree	MST theory				
	Disjoint Set				
	Find the MST weight				
Hard Problems	Number of operations to make network connected				
	Accounts merge				
	Number of islands II				
	Making a large island				
	Most stones removed with same row or column				
	Kosaraju's algorithm				
	Bridges in graph				
	Articulation point in graph				
Dynamic Programming	Climbing stairs				
	Frog Jump				
	Frog jump with K distances				
	Maximum sum of non adjacent elements				
	House robber				
	Ninja's training				
DP on Grids	Grid unique paths				
	Unique paths II				
	Minimum Falling Path Sum				
	Triangle				
	Cherry pickup II				
DP on Stocks	Best time to buy and sell stock				
	Best time to buy and sell stock II				
	Best time to buy and sell stock III				
	Best time to buy and sell stock IV				
	Best time to buy and sell stock with transaction fees				
DP on subsequences	Subset sum equals to target				
	Partition equal subset sum				
	Partition a set into two subsets with minimum absolute sum difference				
	Count subsets with sum K				
	Count partitions with given difference				
	0 and 1 Knapsack				
	Minimum coins				
	Target sum				
	Coin change II				
	Unbounded knapsack				
	Rod cutting problem				
LIS	Longest Increasing Subsequence				
	Print Longest Increasing Subsequence				
	Largest Divisible Subset				

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LIS	Longest String Chain				
	Longest Bitonic Subsequence				
	Number of Longest Increasing Subsequences				
DP on strings	Longest common subsequence				
	Longest common substring				
	Longest palindromic subsequence				
	Minimum insertions to make string palindrome				
	Minimum insertions or deletions to convert string A to B				
	Shortest common supersequence				
	Distinct subsequences				
	Edit distance				
	Wildcard matching				
MCM DP	Matrix chain multiplication				
	Minimum cost to cut the stick				
	Burst balloons				
	Palindrome partitioning II				
Tries	Trie Implementation and Operations				
	Trie Implementation and Advanced Operations				
	Longest Word with All Prefixes				
	Number of distinct substrings in a string				
	Maximum XOR of two numbers in an array				
	Maximum Xor with an element from an array				
Strings(Advanced)	Reverse every word in a string				
	Minimum number of bracket reversals to make an expression balanced				
	Count and say				
	Rabin Karp Algorithm				
	Z function				
	KMP Algorithm or LPS array				
	Shortest Palindrome				
	Longest happy prefix				
Maths	Print all primes till N				
	Prime factorisation of a Number				
	Count primes in range L to R				